IMU & Camera-based Bicycle Crash Detection

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Background and Motivation

Traffic accident detection systems have traditionally focused on motor vehicles, where embedded sensors, airbags, and onboard computing enable Automatic Crash Notification and Advanced Automatic Crash Notification systems. Commercial solutions such as OnStar, Ford 911 Assist, and BMW Assist demonstrate that rapid detection and automatic emergency notification significantly reduce fatality rates by shortening emergency response times. Even a small reduction in response time can reduce accident fatalities by measurable margins. However, these systems share three fundamental limitations:

1. They are expensive, requiring specialized hardware and subscription services.
2. They are vehicle-dependent, embedded directly into engineered, powered vehicles.
3. They are inaccessible to non-motorized road users, particularly cyclists.

Problem Statement

Cyclists, particularly those riding alone on isolated roads, trails, or rural routes are uniquely vulnerable to serious injury or death following a crash due to delayed emergency response!

Unlike motor vehicle occupants, cyclists:

- Lack embedded crash detection systems
- Often ride without witnesses
- Frequently crash in environments with poor visibility or limited cellular activity
- Are at high risk of incapacitation (loss of consciousness, severe injury, immobility)

When a solo cycling accident occurs and the rider is incapacitated, there is often no mechanism to automatically request help. Emergency response is delayed until a passerby discovers the victim if one arrives at all. These delays can turn survivable injuries into fatal outcomes.

User and Purchaser

The main users are urban commuters and mountain bikers who often ride alone or at night. Commuters need protection in high-traffic areas where drivers might not see them, while mountain bikers need a safety line on remote trails where a crash could leave them stranded. Both groups need a device that works passively without interrupting their ride.

We envision two main types of purchasers. The first are the riders themselves, who want the data and legal evidence if something goes wrong. The second, and likely larger group, consists of parents and partners. These buyers are motivated by worry; they are purchasing the device to ensure a loved one’s safety, effectively buying peace of mind for when their partner or child is out riding solo.

State of the Art

Current safety and incident-response technologies can be found in automotive systems, satellite devices, and smartphone applications. While these solutions provide partial functionality, none offer a comprehensive, bicycle-specific “black box” system that integrates crash detection, video evidence, data logging, and automatic emergency notification.

Existing Commercial Products

Automotive eCall Systems are widely deployed in modern passenger vehicles. They automatically detect severe crashes and transmit location and vehicle data directly to emergency responders. While highly effective for cars, automotive eCall systems are completely unavailable for bicycles due to their reliance on vehicle-integrated sensors, power systems, and cellular modules.

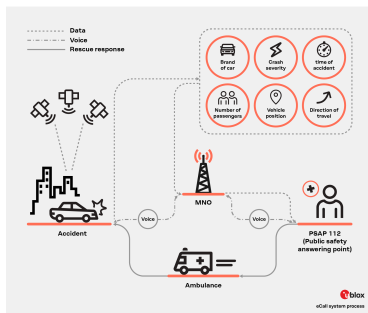

Figure 1. Automotive eCall Systems

Satellite Trackers, such as Garmin inReach, provide emergency SOS capabilities and global location tracking. These devices are primarily designed for outdoor recreation and remote expeditions rather than urban or daily cycling. They require expensive standalone hardware and ongoing subscription fees, making them impractical for daily bicycle commuting. Additionally, they do not continuously record sensor data or video footage leading up to an accident.

Figure 2. Garmin inReach Satellite Tracker

Smartphone Applications, including Strava Beacon and Apple Crash Detection, detect crashes using onboard phone sensors. These solutions suffer from high false-positive rates, phone dropped as an example, and depend on phone placement and orientation. In the event of a serious crash, the smartphone may be damaged or lose power, preventing SOS transmission. Smartphones are not optimized for continuous video recording and storage of incident data.

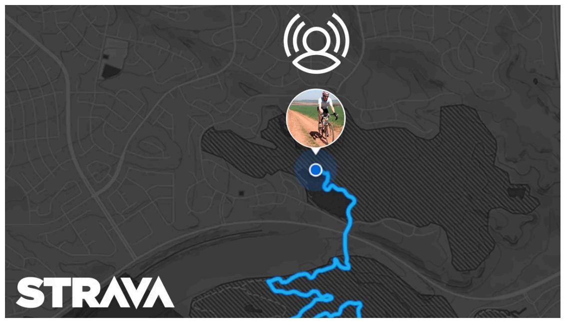

Figure 3. Strava Beacon smartphone app

Existing Patents

The first patent (US10144474B2) we researched detects collisions by monitoring abrupt changes in acceleration, velocity, or force using onboard sensors. When measured values exceed predefined thresholds, the system determines that a collision has occurred. The detection logic is optimized for vehicles with predictable motion patterns and sufficient mass to distinguish crashes from normal operation.

This device is designed for automotive applications and does not account for the high vibration and dynamic variability present in bicycles. It does not include cyclist-specific emergency notification capabilities. Additionally, it lacks contextual data such as video footage or detailed motion logs.

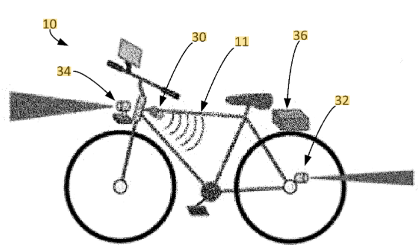

Figure 4. Collision detection, US10144474B2

Another patent of our interest is an accident detection and notification system integrated into a motorcycle's electrical and sensor framework. It monitors parameters such as vehicle speed, tilt angle, and impact forces to determine whether a crash has occurred. Upon detection, the system transmits accident information, including location data, to a remote server or emergency contact. The design assumes continuous access to vehicle power and fixed sensor placement.

However, the system is specific to motorcycles and depends on engine-based sensors and electrical systems not present on bicycles. Its detection method is not usable to human-powered vehicles. Furthermore, it does not include video recording or detailed pre-crash data storage.

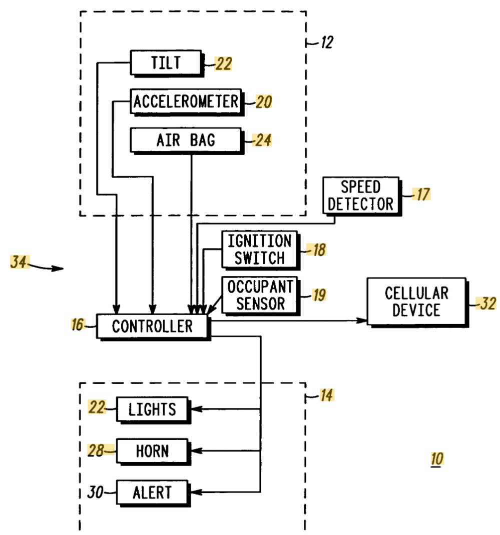

Figure 5: Motorcycle accident notification system.

The final preexisting patent (US7129826B2) focuses on notifying nearby users or devices when an accident is detected within a localized area using short-range communication methods to broadcast an alert following a detected collision event.

The major drawback is that it does not automatically contact emergency responders, relying instead on nearby individuals to take action. This approach is ineffective for solo cyclists in remote or low-traffic areas.

Together, these existing solutions fail to address the need for a robust, bicycle-specific system that autonomously detects crashes, preserves evidence, and initiates emergency response.

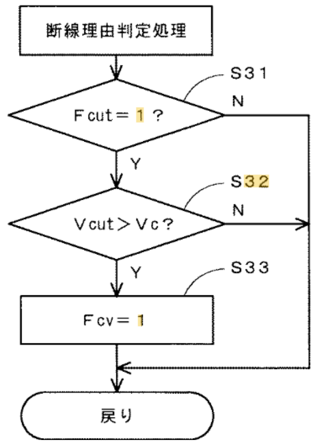

Figure 6: Localized accident notification, US7129826B2

Specifications

Design priorities include detection accuracy, reaction speed, and durability. To distinguish crashes from hard braking, or a pothole, the system requires a false positive rate below 10%. Reaction time must be under 10 seconds to address medical emergencies quickly. As a "black box," the enclosure must survive impacts to preserve data, while the battery must support at least 6 hours of operation for long rides.

Economic and safety constraints further scope the project. Material costs are capped at $60 USD, and the installation must never obstruct steering or braking. We enforce a local-only storage policy for user privacy. To ensure video is legally useful, the camera requires 1080p resolution and a 120-degree field of view. Finally, a physical cancel button with a 15 to 30 second grace period handles false alarms like a tipped bike.

Specification	Justification	Quantification
Detection Accuracy	System must distinguish crashes from sharp stops	False Positive Rate: < 10%
Response Time	In medical emergencies time is very important	Trigger Time: < 10 seconds
Power	Must last for a standard long distance ride	Battery Life: > 4 hours
Durability	Device must survive large forces to save data	Survive crash test

Visual Evidence	Should capture clear footage of the incident and surroundings to be able to prove liability	Resolution: ≥ 720 @ 15fps Field of View: ≥ 120° (Wide Angle)
False Alarm Prevention	User must be able to cancel the emergency call if the crash detection was a mistake	Grace Period: 15–30 seconds Interface: 1 Physical Cancel Button
Cost	Must be priced comparable to market	Cost: < $150 USD
Safe	The device cannot interfere with the rider's ability to ride the bike	No interference with steering/brakes.
Legal	Must comply with US laws for contacting help	Most likely connect to the users phone to call

Figure 7. Specifications Table for Bike Crash Detection

Solutions & Alternatives Matrix

Our team came up with a wide range of solutions for the larger problem of unmonitored bike crashes, specifically in two categories: impact prevention and crash detection/response.

Solutions in the first category primarily consist of automatic features that help prevent a crash in real time by alerting the rider to hazards. The second category, and the category in which our final solution falls, revolves around methods to detect crashes and respond to them by notifying emergency services.

To assess which solutions were best, we created a weighted matrix where each solution's specifications were rated on a scale of 1 (poor) to 5 (great). Specifications were weighted based on importance: accuracy and feasibility received the highest weights (5 each), followed by data quality (4), then ease of use (4), with cost and durability weighted at 3 each.

Solutions	Accuracy	Feasibility	Data Quality	Ease of Use	Cost	Durability	Total
Smart Sensor Box (Selected)	5	4	5	4	5	4	**108**
Ball Switch Tilt Array	4	5	3	4	4	5	**100**
Break-Wire Separation Detector	5	4	2	2	5	3	**85**
Ultrasonic Rear Proximity	3	3	3	4	2	2	**70**

Warning							
Wheel Speed Sensor w/ Auto Braking	4	2	4	3	2	2	**70**

Figure 8. Solutions & Alternatives Matrix; Weighting: Accuracy & Feasibility (5), Data Quality & Ease of Use (4), Cost & Durability (3)

Prevention Solutions

We proposed two prevention-focused solutions that alert riders to hazards before a crash occurs. The first is an ultrasonic rear proximity warning system, which calculates the speed of approaching vehicles by comparing sequential distance readings, triggering a vibration motor embedded in the handlebars in the case of potential collisions. The goal with this idea was to enable riders to get a sense of their surroundings & obstacles without needing to turn back and be distracted from the road.

Our second prevention idea was a wheel speed sensor with automatic braking. This idea was primarily inspired by Hall effect setups, in which neodymium magnets are attached at equidistant spots along the wheel, and a sensor fixed to one point calculates instantaneous speed from the frequency the magnets pass by at. We tailored this idea to detect two wheel conditions that consistently lead to crashes: wheel lockup, which is when pulses stop but the accelerometer shows continued forward motion above 10 km/h, and rapid deceleration, or speed dropping more than 15 km/h within 500ms. When either condition is detected, the system enters a pre-crash alert state, triggers automatic braking, and slows the rider down.

There were two overarching limitations with both ideas in the prevention category. The first, cost, revolves around the technology needed to power the devices. Both the ultrasonic radar system and the Hall effect setup can range from $100 to $200 for even a DIY setup, and this would make the device unaffordable for most target users, casual solo bikers. The second limitation, is durability, as both systems require technology that is delicate and can be hampered in adverse weather conditions like rain and snow.

Detection Solutions

For crash detection, we evaluated two solutions beyond the smart box that forms our core design. The first alternative is a ball switch tilt array using four ball tilt switches positioned in a custom angled bracket at specific tilts, confirming a crash whenever the tilt state changes from upright to fallen and persists for more than 3 seconds. This solution actually scored second-highest overall because the switches draw essentially zero power, making them extremely efficient and cost friendly, and provide reliable crash detection in proven settings.

Our second detection alternative is a break-wire rider separation detector. A thin wire connects the rider to the bike through a magnetic breakaway connector, creating a closed circuit that is monitored continuously by a microcontroller. If the rider is ever thrown from the bike during a crash, the magnetic connector breaks the circuit, triggering an alert to emergency services. While this solution scored high on accuracy, it ranked lower on ease of use because it requires riders to attach the tether before each ride.

For our selected solution, we decided that implementing an IMU-based smart sensor box would offer the best solution for our core crash detection system. Specifically, we plan on using a MPU-6050 IMU unit which combines an accelerometer and gyroscope in a single package, supported by a rear-facing camera for video detection of incoming obstacles. When the accelerometer and gyrometer detects irregular force, or if the camera detects rapidly approaching objects, the system triggers a 30-second countdown. If the rider does not cancel the alert by pressing a button, an SOS message with GPS coordinates is sent to emergency contacts via Bluetooth connection to the rider's smartphone.

Timeline and Plan

Week 4	Prototype Design and Materials Purchase
Week 5	Initial Prototype
Week 6	Initial Testing (RC Crashes)
Week 7	Finalize Prototype Functionality
Week 8	Final Testing On Users (Dartmouth Biking Club)
Week 9 & 10	Final Presentation & Report

Roles & Responsibilities

Victory Igwe (Project Manager): Oversee the team progress, ensure the timeline is followed, coordinate meetings with instructors, and manage project documentation and deliverables.

Mason Shriver (Treasurer / Manufacturing Engineer): Manage the budget, order electronic components (IMU, etc), and lead the physical assembly and prototype of the circuitry.

Aniket Dey (Software & Algorithms Lead): Research and implement the crash detection logic, develop the video loop recording script, and handle the Bluetooth SOS integration.

Julian Vu (Design & Testing Engineer): Lead the CAD modeling of the weather-resistant enclosure and mounting system, develop testing procedures, and coordinate crash simulations to verify performance.

References:

A. "eCall System Process Diagram." *u-blox*, u-blox AG, 2024, content.u-blox.com/sites/default/files/2024-02/eCall-5+.png.

B. "Garmin inReach Explorer." *HikingGuy*, hikingguy.com/wp-content/uploads/garmin-inreach-explorer-featured.jpg.

C. "Strava Beacon Map Feature." *BikeRadar*, Immediate Media Company, 2019, c02.purpledshub.com/uploads/sites/39/2019/03/strava-beacon_map-1470306152054-se724iu2ek0x-12d57c3.jpg.

D. Nitz, Karl, et al. *Localized Accident Notification*. US7129826B2, United States Patent and Trademark Office, 31 Oct. 2006. Google Patents, patents.google.com/patent/US7129826B2/en.

E. Rajamani, Rajesh, and Woongsun Jeon. *Collision Detection*. US10144474B2, United States Patent and Trademark Office, 4 Dec. 2018. Google Patents, patents.google.com/patent/US10144474B2/en.

F. *Motorcycle Accident Notification System*. JP4345948B2, Japan Patent Office. Google Patents, patents.google.com/patent/JP4345948B2/en.